# Project_Deliverable

September 7, 2025

## 1 Project Deliverable #1

Nebraska Tractor Tests (Dataset on Brightspace)

Description: Explore the impacts of draft, and Power Take-Off (PTO) for different types of tractors for various models and years. Using this dataset will require that you look up and learn more about the different tractor models, in order to ask reasonable research questions.

[15]:
```
import pandas as pd
import numpy as np

df = pd.read_excel('../Data/12_Nebraska_Tractor_Tests_data.xlsx')
```

### 1.1 A detailed dataset description

1. Meta-data: Include key information like file formats, structure, data types (e.g. tabular, image) measurement formats (e.g. categorical, numeric) units, granularity, dimensions, and any temporal or geographic details.
2. Variables: Identify and describe response and explanatory variables for each research question

[8]:
```
df
```

[8]:
```
         Id               Pdf Link  Year Nebraska Test Number  \
0         1  Click for test report  1920                     1
1         2  Click for test report  1920                     2
2         3  Click for test report  1920                     3
3         4  Click for test report  1920                     4
4         5  Click for test report  1920                     5
…     …                        …   …                      …
3841   3842  Click for test report  2022                  2257
3842   3843  Click for test report  2022                  2258
3843   3844  Click for test report  2022                  2259
3844   3845  Click for test report  2022                  2260
3845   3846  Click for test report  2022                  2261

     Nebraska Summary Number             Manufacturer          Make     Model  \
0                        NaN  Waterloo Gas Engine Co.  Waterloo Boy  N, 12-25
1                        NaN                      NaN     Withdrawn       NaN
2                        NaN                J.I. Case          Case     10-18
```

```
3                        NaN                J.I. Case          Case     15-27
4                        NaN                J.I. Case          Case     22-40
…                        …                    …              …        …
3841                   S1220                John Deere     John Deere    5125M
3842                   S1221                John Deere     John Deere  9R 390
3843                   S1222                John Deere     John Deere  9R 440
3844                   S1223                John Deere     John Deere  9R 490
3845                   S1224                John Deere     John Deere   6140M

     Transmission Chassis Fuel  PTO HP  Belt HP  Drawbar HP            Remarks
0             MAN      2WD    K     NaN     25.0        15.0               NaN
1             NaN      NaN  NaN     NaN      NaN         NaN               NaN
2             MAN      2WD    K     NaN     18.0        11.0               NaN
3             MAN      2WD    K     NaN     31.0        21.0               NaN
4             MAN      2WD    K     NaN     49.0        31.0               NaN
…           …      …   …      …       …         …                  …
3841      PRPS 4      FWA    D   109.0      NaN       100.0               NaN
3842         PSH      4WD    D   323.0      NaN       296.0               NaN
3843         PSH      4WD    D   321.0      NaN       340.0  partial power PTO
3844         PSH      4WD    D   321.0      NaN       382.0  partial power PTO
3845      PRPS 4      FWA    D   112.0      NaN       110.0               NaN

[3846 rows x 15 columns]
```

[7]:
```
df.dtypes
```

[7]:
```
Id                          int64
Pdf Link                   object
Year                        int64
Nebraska Test Number       object
Nebraska Summary Number    object
Manufacturer               object
Make                       object
Model                      object
Transmission               object
Chassis                    object
Fuel                       object
PTO HP                    float64
Belt HP                   float64
Drawbar HP                float64
Remarks                    object
dtype: object
```

[10]:
```
df['Pdf Link'].unique()
```

[10]:
```
array(['Click for test report'], dtype=object)
```

[12]:
```
df['Nebraska Test Number'].unique()
```

[12]:
```
array(['1', '2', '3', …, '2259', '2260', '2261'],
      shape=(2356,), dtype=object)
```

[22]:
```
np.where(df['Nebraska Test Number'].str[:7]=='Special')
```

[22]:
```
(array([89]),)
```

[27]:
```
df[89:90]
```

[27]:

| | Id | Pdf Link | Year | Nebraska Test Number | Nebraska Summary Number | Manufacturer | Make | Model | Transmission | Chassis | Fuel | PTO HP | Belt HP | Drawbar HP | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 89 | 90 | Click for test report | 1921 | Special A | NaN | Shaw-Enochs | Shawnee | POWER PATROL | MAN | 2WD | G | NaN | NaN | 7.0 | Tested no Enochs Shawnee 685, no PTO |

[29]:
```
len(df['Remarks'].unique())
```

[29]:
```
653
```

[36]:
```
len(df['Nebraska Summary Number'].unique())
```

[36]:
```
1326
```

[35]:
```
df['Fuel'].value_counts(dropna=False,normalize=True)
```

[35]:
```
Fuel
D       0.765211
G       0.143266
K       0.028861
LPG     0.026781
DIST    0.020021
NaN     0.010920
TF      0.004940
Name: proportion, dtype: float64
```

[37]:
```
len(df['Make'].unique())
```

[37]:
```
143
```

[38]:
```
len(df['Model'].unique())
```

[38]:
```
2484
```

[39]:
```
len(df['Transmission'].unique())
```

[39]:
```
18
```

[40]:
```
len(df['Chassis'].unique())
```

[40]:
```
10
```

[41]:
```
df['Chassis'].value_counts(dropna=False,normalize=True)
```

[41]:
```
Chassis
2WD             0.438638
FWA             0.381175
Crawler         0.079563
4WD             0.078263
NaN             0.010920
4TD             0.007280
half track      0.001560
See Remarks     0.001300
FTA             0.001040
Single Wheel    0.000260
Name: proportion, dtype: float64
```

[11]:
```
df.describe()
```

[11]:

| | Id | Year | PTO HP | Belt HP | Drawbar HP |
|---|---|---|---|---|---|
| count | 3846.000000 | 3846.000000 | 2953.000000 | 752.000000 | 3633.000000 |
| mean | 1923.500000 | 1981.868175 | 128.743312 | 40.240691 | 101.293614 |
| std | 1110.388896 | 26.641834 | 93.554361 | 20.497846 | 88.677504 |
| min | 1.000000 | 1920.000000 | 11.000000 | 1.000000 | 0.600000 |
| 25% | 962.250000 | 1962.000000 | 63.000000 | 27.000000 | 41.000000 |
| 50% | 1923.500000 | 1984.000000 | 99.000000 | 36.000000 | 71.000000 |
| 75% | 2884.750000 | 2005.000000 | 164.000000 | 49.000000 | 131.000000 |
| max | 3846.000000 | 2022.000000 | 598.000000 | 145.000000 | 531.000000 |

[34]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Create a figure with a 1x2 grid of subplots
fig, axes = plt.subplots(1, 3, figsize=(10, 5))

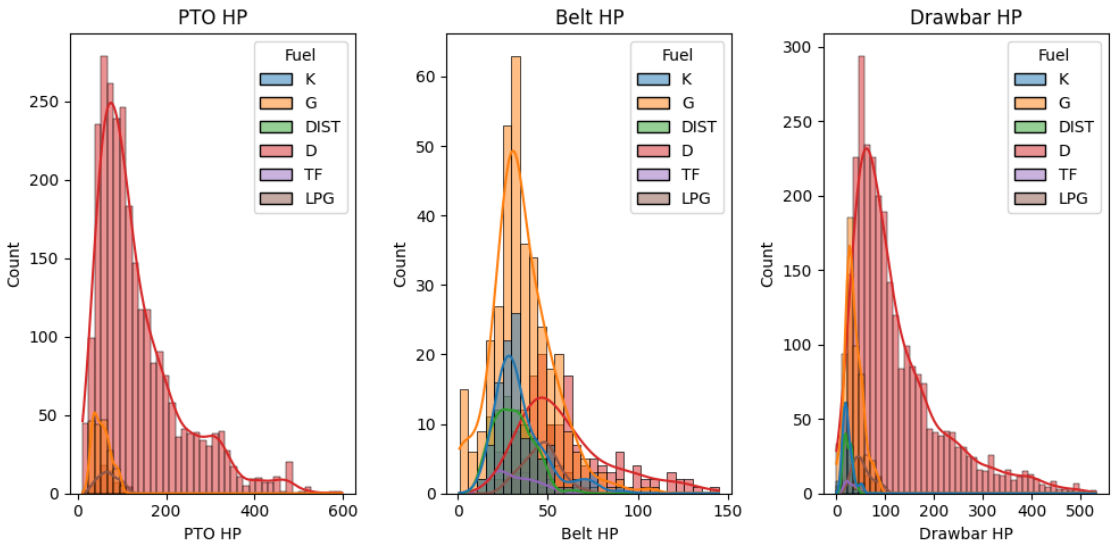
# Plot on the first subplot (axes[0])
sns.histplot(x="PTO HP", data=df, ax=axes[0], hue='Fuel', kde=True)
axes[0].set_title('PTO HP')

# Plot on the second subplot (axes[1])
```

```
sns.histplot(x="Belt HP", data=df, ax=axes[1], hue='Fuel', kde=True)
axes[1].set_title('Belt HP')

# Plot on the second subplot (axes[1])
sns.histplot(x="Drawbar HP", data=df, ax=axes[2], hue='Fuel', kde=True)
axes[2].set_title('Drawbar HP')

plt.tight_layout() # Adjust layout to prevent overlapping
plt.show()
```




The dataset is compose of 3 variables related to Horse Power:

- PTO HP: Horsepower Power Take Off - type float
- Belt HP: the power transmitted by a V-belt from a driving pulley to a driven pulley - type float
- Drawbar HP: the usable power available at the drawbar or rear of a tractor, or a locomotive, to pull a load or implement - type float

## 1.2 Three research questions you have about the data

Hint: For each question, try to identify the explanatory variable(s) (the predictors) and the response variable (the outcome you're trying to explain). This step is important because clearly linking predictors and outcomes helps you choose the right statistical methods later and ensures your questions can actually be answered with the data available.

Based on the review of the data, i have selected that the most interesting feature to predict is Drawbar HP, based on the PTO HP, Belt HP, the kind of Gear, fuel, transmission and chasis. With this information the 3 questions I would like to answer with the data are the following:

1. Considering a fixed kind of fuel, which are the most important variables to predict the amount of Drawbar HP?

2. Are there any temporal pattern in the evolution of the Drawbar HP?
3. Are there any other confounding variables that might predict the amount of Drawbar HP? Consider analysing the brand and model, to see if there is a certain bias, based on the manufacturer.